IT PROJECT 2 | SARAWAK TUN OPENG UNIVERSITY OF TECHNOLOGY

Web-Based Preorder and Freshness Management System for Home-Based Entrepreneurs

CASE STUDY: WRAP & BOWL KUCH · KUCHING, SARAWAK

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HOW THE SYSTEM CONNECTS THE WORK

01 **Browse food** → 02 **Choose date and area** → 03 **Check total and stock** → 04 **Save order and deduct stock** → 05 **Review orders and sales**

01 Background

- Preorder details can be scattered across messages.
- Food-safety concerns motivate clearer ingredient storage records.
- Reviewed platforms provide limited control over preordering and independent branding.

02 Original objectives

1. To design and develop a web application that supports flexible date-based food preordering and scheduling for home-based food entrepreneurs.

2. To implement a food stock freshness monitoring feature based on ingredient shelf life and storage duration.

3. To develop a personalised single-vendor storefront module that allows sellers to strengthen branding without direct competitor distraction.

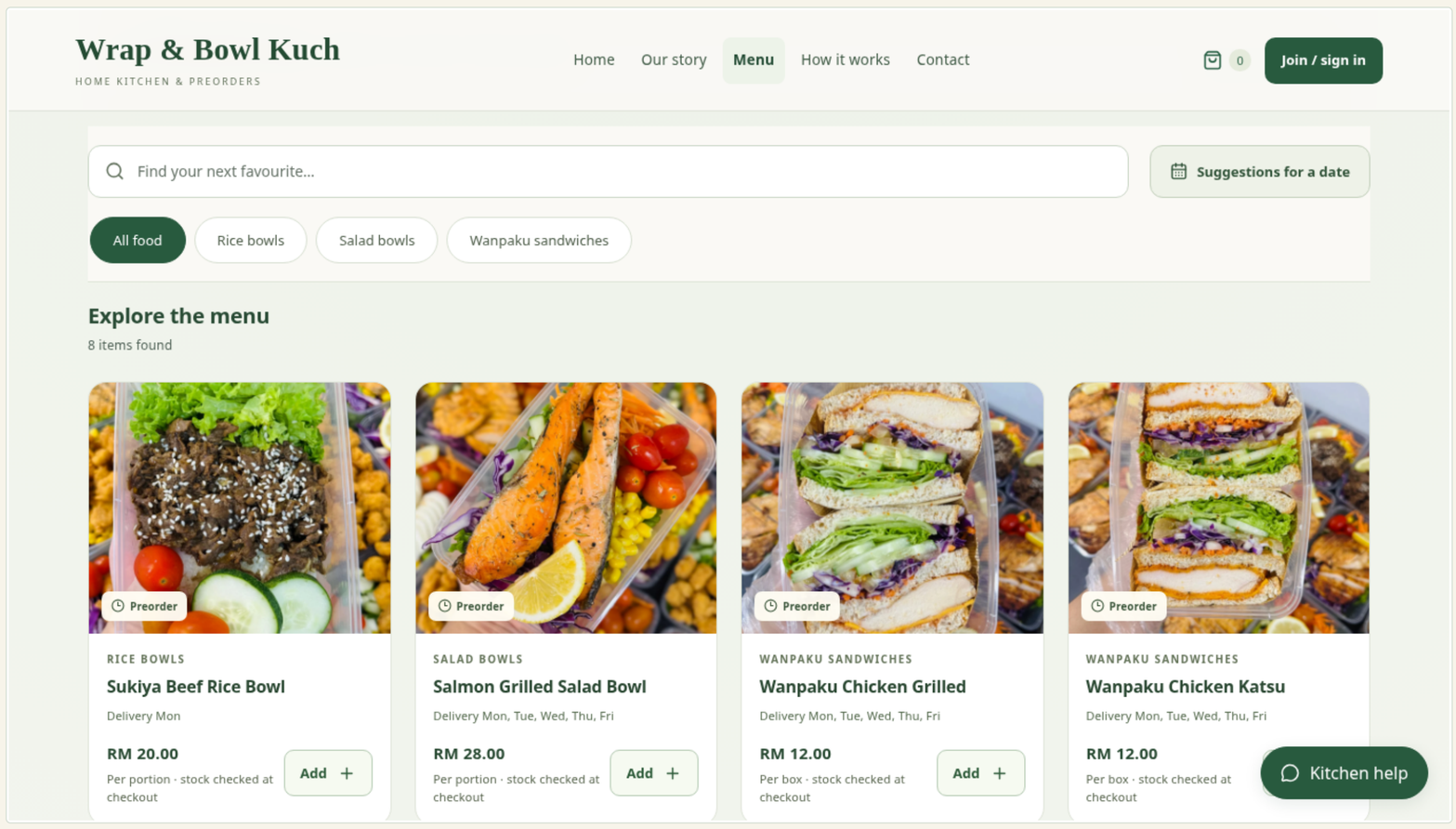
4. To implement a sales analytics dashboard that supports data-driven business decision-making for home-based food operators.

5. To develop a menu recommendation system that reduces customer decision fatigue and improves purchase conversion rates.

03 Development approach

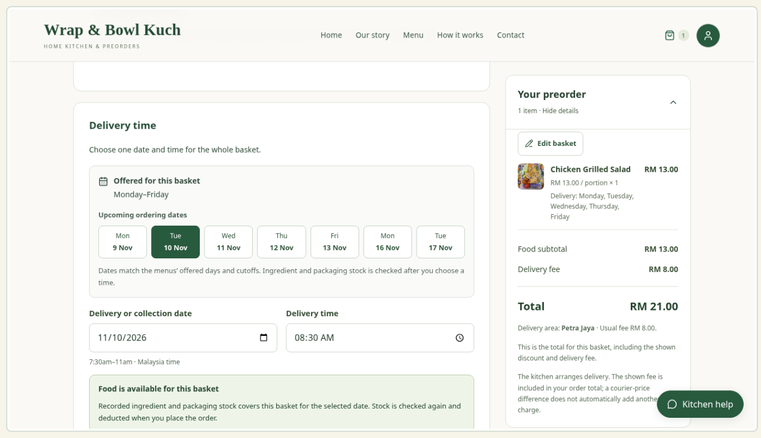
- Review literature and client needs.
- Build and refine the application from client feedback.
- Check functions; collect formal user feedback next.

04 Working system



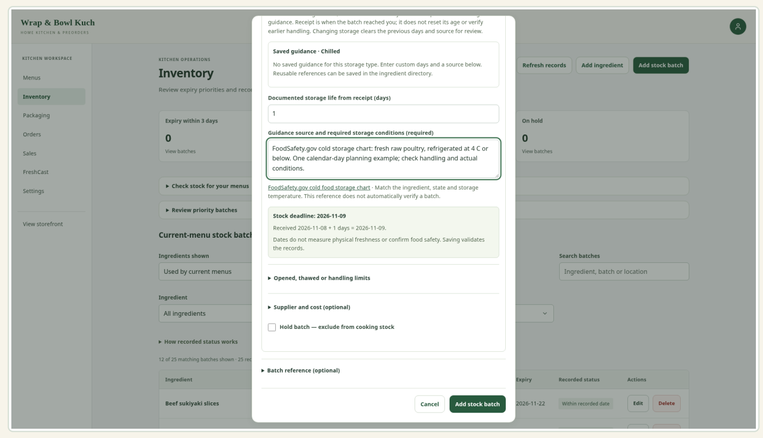
Client menu photos, prices and selling units in one branded storefront.

05 Connected preorder and stock records



Preorder and stock deduction

Checkout checks an offered date and total. Order placement deducts eligible ingredients and packaging together.



Recorded storage life

The owner records date evidence and storage life. Earlier deadlines and holds affect stock allocation.

IMPLEMENTATION

React	Django	Supabase / PostgreSQL
Customer and owner screens	Pricing, access and stock rules	Persistent application records

06 Verification

One recorded preorder changed:

Chicken: 5,000 g → 4,870 g
Salad bowls: 200 → 199

Observed in the local functional test.

Conclusion & next step

The prototype connects preorders, recorded stock life and sales information.

Formal client and student evaluation is planned.

Explore the menu

Scan to open the live storefront.

Public menu
Not a payment QR

Screens: local application, client photos and example records. Dates support planning; they do not guarantee food safety.

Selected references: Pallianysamy et al. (2025), Home-based Foods in Malaysia; FoodSafety.gov, Cold Food Storage Chart.